

GANDHIMATHI (MATHI) PADMANABAN

Ph.D. (Aug 2026) | University of Michigan-Dearborn

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RESEARCH PROFILE

Doctoral researcher in applied computer vision and machine learning. I build perception that human-AI systems can rely on: it keeps working when the camera, platform, or site changes, and it abstains rather than report a confident error. I ground the measurement in projective geometry, using it to derive features and to reject detections that are physically impossible. In my dissertation I measured vehicle-bicyclist interactions from a single uncalibrated bicycle camera; the same geometric formulation, retrained on the Waymo Open Dataset with LiDAR supervision, reached $R^2 = 0.812$ on a different platform and range. A second line models driver behavior from vehicle kinematics. I want to carry this approach, building the real structure of a problem into the model, into new domains where cheap sensors must make reliable measurements.

EDUCATION

- Aug 2026 **Ph.D., Industrial and Systems Engineering** - *University of Michigan-Dearborn, USA*
Dissertation: “Adaptable Machine Learning and Computer Vision Frameworks for Road Safety: Applications to Driver Behavior and Driver-Bicyclist Interaction Research” (*defended June 4, 2026*)
Advisor: [Dr. Fred Feng](#)
- 2021 **M.S., Human Centered Design and Engineering** - *University of Michigan-Dearborn, USA*
Thesis: “[Computational Human Performance Modeling using Queuing Network in an Open-Source Platform](#)”
Advisor: [Dr. Fred Feng](#)
- 2013 **B.E., Computer Science and Engineering** - *Anna University, India*
Thesis: “Automated Detection of Modifications in Software Requirement Traceability Links”
Advisor: Prof. Ramachandran Alagarsamy
- Certifications:** Connected & Automated Transportation Certificate (*Aug 2026*); Rackham DEI Certificate (2025); Post Graduate Diploma in Computer Applications (2011)

RESEARCH EXPERIENCE

- 2022–2026 **Graduate Student Research Assistant** - *University of Michigan-Dearborn, USA*
Advisor/PI: [Dr. Fred Feng](#)
- Designed and implemented an **experimental platform for driver-bicyclist interactions** with integrated RGB camera and LiDAR sensing, achieving 98.1% recall with 1 false positive on 315 annotated events.
 - Built a **fine-grained vehicle body-type classifier** by fine-tuning a Vision Transformer (ViT) on a custom dataset I assembled, with categories defined by injury risk to cyclists; held 89% accuracy on held-out sites with no retraining, and designed it to abstain under low confidence rather than misclassify.
 - Designed a calibration-free lateral passing distance estimation system using projective geometry-derived bounding box features (0.126 m MAE; external validation on Waymo Open Dataset: $R^2=0.812$), showing the geometry-grounded feature design transfers across sensing platforms.
 - Led manuscript preparation across multiple venues (2 peer-reviewed, 4 under review and preprints).
 - Mentored undergraduate and master’s students in computer vision and Python data pipelines, including contributions to open-source tooling.
- 2011–13 **Research Assistant** - *Anna University, India*
Advisor/PI: Prof. Ramachandran Alagarsamy
- Contributed to software testing research group; led development of university applications including exam cell software, attendance management software and sports management application using full-stack development approaches.
 - Developed automated detection system for modifications in software requirement traceability links (Bachelor’s thesis)

PUBLICATIONS

Peer-Reviewed Works

1. [Gandhimathi Padmanaban](#), Fred Feng, Edward Dai, Ankit Saini, Guopeng Hu, and Yanan Zhao. “A Comparative Analysis of Acceleration and Deceleration Profiles for Aggressive Driving Styles and Fuel Economy Test Cycles”. *WCX SAE World Congress Experience*. 2025. DOI: <http://dx.doi.org/10.4271/2025-01-8605>.

2. **Gandhimathi Padmanaban**, Nathaniel P. Jachim, Hala Shandi, Lilit Avetisyan, Garrett Smith, Howrah Ham-moud, and Feng Zhou. “An Autonomous Driving System - Dedicated Vehicle for People with ASD and their Caregivers”. *AutomotiveUI '21 Adjunct: 13th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. Association for Computing Machinery, 2021, pp. 142–147. DOI: <https://doi.org/10.1145/3473682.3480282>.

Papers Under Review & Preprints

1. **Gandhimathi Padmanaban**, Fred Feng, Edward Dai, Ankit Saini, Guopeng Hu, and Yanan Zhao. “A Machine Learning Framework to Identify Aggressive Driving Based on Vehicle Kinematics and Driver Pedal Operations”. *Under Review — Engineering Applications of Artificial Intelligence*. 2025.
2. **Gandhimathi Padmanaban**, Rayane Moustafa, and Fred Feng. “A Geometry-Informed Computer Vision Method for Detecting and Examining Overtaking Vehicles From A Bicycle”. *In Preparation for Accident Analysis and Prevention*. 2026. URL: <https://par.nsf.gov/biblio/10679667>.
3. **Gandhimathi Padmanaban** and Fred Feng. “Vision-Based Lateral Passing Distance Estimation from Bicycle-Mounted Cameras: A Projective Geometry Approach”. *Submitted to International Cycling Safety Conference*. 2026.
4. **Gandhimathi Padmanaban** and Fred Feng. “An Open-Source Two-Stage Computer Vision Pipeline for Fine-Grained Vehicle Classification using Vision Transformers”. *arXiv preprint; arXiv:2606.05149*. 2026. DOI: <https://doi.org/10.48550/arXiv.2606.05149>.

TECHNICAL EXPERTISE

<i>Programming</i>	Expert: Python (TensorFlow, PyTorch, scikit-learn, OpenCV, pandas), Git, L ^A T _E X; Proficient: Julia, R, MATLAB, C#, UI Frameworks, SQL; HPC: Great Lakes cluster (Slurm)
<i>AI/ML Methods</i>	Geometry-Informed Computer Vision, Deep Learning (CNN, RNN, Transformers), Supervised Learning, Ensemble Methods, Sequential and dynamical system modeling, Bayesian Methods, Out-of-Distribution Detection.
<i>Autonomous Systems</i>	Sensor fusion (camera + LiDAR), perception pipelines, on-road sensing platform design, vehicle kinematics modeling, ROS (familiar)
<i>Tools & Frameworks</i>	OpenCV, YOLOv5, RT-DETR, ByteTrack, Multi-Object Detection & Tracking, Multi-Modal Data Fusion, LiDAR
<i>Foundations</i>	Optimization, statistical learning, multivariable calculus, mathematical modeling, geometric computer vision, vision transformers, Vision-Language Models (familiar), dynamical systems (geometry and stability)
<i>Research</i>	Experimental Design, Statistical Analysis, Scientific Writing, Peer Review, Institutional Review Board (IRB) Protocols, Cross-Validation, Academic Collaboration
<i>Domains</i>	Transportation Safety, Naturalistic Driving Studies, Driver Behavior Analysis, Driver–Bicyclist Interactions, Autonomous & Connected Vehicles, Human Factors, Safety-Critical Perception

AWARDS AND GRANTS

- **Student Visionary Award** - International Forum on Research Excellence Conference (IFoRE' 25), Sigma Xi-The Scientific Research Honor Society, 2025
- **Upsilon Pi Epsilon (UPE) Scholarship** - Awarded for exceptional academic performance, extracurricular involvement, and leadership within the computing community, 2024
- **Global Finalist - NASA Space Apps Challenge**, 2023
- **Irma M. Wyman Scholar**, *Center for the Education of Women (CEW+)*, *University of Michigan*, 2020-2021 (\$11,500)
- Non-Resident Graduate Student Scholar, *University of Michigan-Dearborn*, 2020-2021 (\$13,000)
- Deloitte Hackathon Special Mention, 2017 | Syncfusion Hackathon 2nd Place, 2015 (INR 35,000)

TALKS AND PRESENTATIONS

Talks

- **WocCode – University of Michigan Ann Arbor**, Feb 2024
NASA SpaceApps Experience: DigitwiML – Open-Source Project to model Digital Twin of C.elegans in Space
- **University of Michigan–Dearborn**, Sep 2021
Guest Talk: IMSE 501 Human Factors & Ergonomics

Conference Presentations

- **Transportation Research Board Annual Meeting**, Jan 2026 (*Poster*)
A Geometry-Informed Computer Vision Method for Detecting and Examining Overtaking Vehicles From A Bicycle
- **Transportation Research Board Annual Meeting**, Jan 2026 (*Presentation*)
Quantifying Drivers-Overtaking-Bicyclists with Surrogate Safety Measures Derived from High-Resolution Digital Lidar (Moustafa, Padmanaban, Feng)
- **International Forum on Research Excellence (IFoRE' 25)**, Oct 2025 (*Workshop Session*)
Hybrid Physics-Data Modeling for Sustainable Transportation: Bridging Classical Models and Modern AI
- **WCX SAE World Congress Experience**, Apr 2025 (*Paper Presentation*)
A Comparative Analysis of Acceleration and Deceleration Profiles for Aggressive Driving Styles and Fuel Economy Test Cycles
- **Automotive User Interfaces (AutoUI) Conference**, Sep 2021 (*Paper Presentation*)
An Autonomous Driving System – Dedicated Vehicle for People with ASD and their Caregivers

RESEARCH SOFTWARE & OPEN SOURCE PROJECTS

- **DigitwiML** (NASA SpaceApps Global Finalist, 2023): Machine learning platform for digital twin modeling of biological dynamics in space environments. First application of data-driven dynamical modeling beyond transportation, demonstrating interest in scientific ML for physical systems - [GitHub](#)
- **Geometry-informed overtaking tracker** - Computer Vision system for tracking vehicle-bicyclist interactions (open-source, Python) - [GitHub](#)
- **Vehicle Type Classifier** - Injury-risk based computer vision classification system for vehicle types (open-source, Python) - [GitHub](#)

PROFESSIONAL EXPERIENCE (INDUSTRY)

Five years building and shipping production software before the PhD, across enterprise consulting and developer-tools product engineering.

- 2017–18 **Development Lead / Consultant**, Deloitte (Offices of the US) – Bangalore, India
- 2015–17 **Programmer Analyst**, Cognizant – Chennai, India
- 2013–15 **Software Engineer Level-II**, Syncfusion – Chennai, India

TEACHING AND MENTORING

- 2026 **Teaching Assistant** - *Python as an Engineering Tool*, University of Michigan-Dearborn
Intensive Python module for visiting engineering students in the ESIAE exchange program.
- 2024–Now **Certified Instructor** - *The Carpentries*
Hands-on computational methods workshops (Python data analysis, Git version control, Unix shell, OpenRefine) for researchers in biosciences, library science, and data-intensive fields.
[University of Tennessee Knoxville \(May 2025\)](#) | [CZI Foundation \(Nov 2024\)](#) | [University of Michigan \(Mar 2024\)](#)
- 2023–Now **Instructor & Mentor** - *WoCCode* - University of Michigan-Ann Arbor
Machine Learning and Python workshop: materials open-sourced. [Machine Learning Workshop \(Feb 2024\)](#) | [Summer 2023 Boot Camp](#)
- 2010–12 **Student Instructor / Teaching Assistant** - Anna University, India
Courses: Artificial Intelligence, Probability and Queuing Theory, Transforms and PDEs, Systems Software Laboratory

SERVICE AND LEADERSHIP

Leadership: President, [Upsilon Pi Epsilon](#) Michigan Beta Chapter (2024-2025) • Member: [SAE International \(AI in Simulation Task Force\)](#) (co-authoring technical guidelines on AI in simulation for ground vehicles), [Sigma Xi](#), [Alpha Pi Mu](#), [Society of Industrial and Applied Mathematics](#), [ACM](#), [HFES](#)

Reviewer: [IEEE-ITS](#), [TRBAM 2026](#), [ICIS 2025](#), [AutoUI 2024](#), [CHI 2024](#), [AMCIS 2024](#), [CUI 2024](#), [IMX 2024](#), [DIS 2024](#)

Community: [MIDAS AI Summit Student Volunteer](#) (2023,2025) • [WoC|CodeNova Hackathon Judge](#) (2025) • [WocCode Mentor](#) • [Carpentries Certified Instructor & Lesson Maintainer](#) • [NASA SpaceApps & MHacks Judge](#) • [STEM High School Tutor](#) (2010-13)